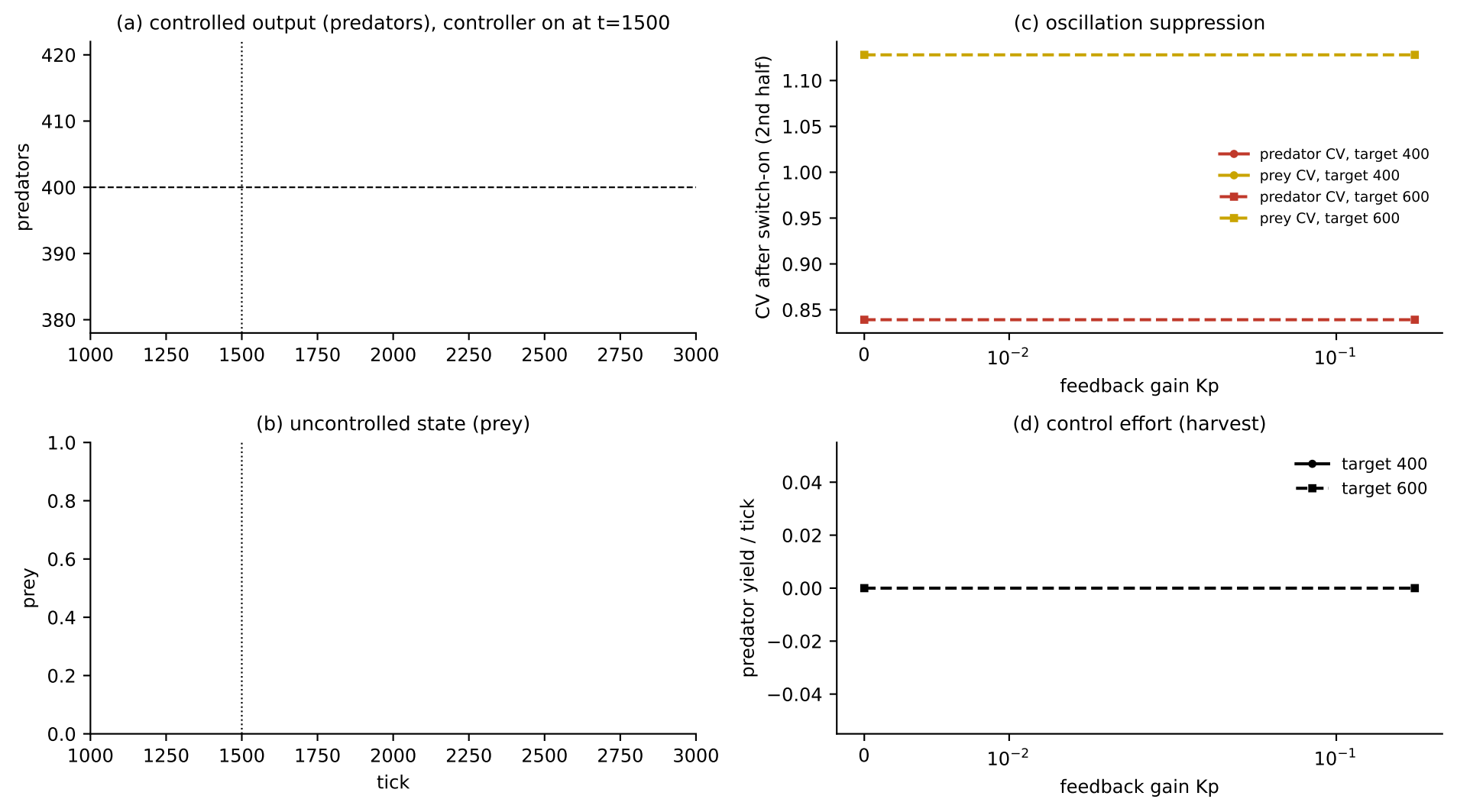
(a) controlled output (predators), controller on at t=1500
predators
420
410
400
390
380
1000
1250
1500
1750
2000
2250
2500
2750
3000
(b) uncontrolled state (prey)
prey
1.0
0.8
0.6
0.4
0.2
0.0
tick
(c) oscillation suppression
CV after switch-on (2nd half)
1.10
1.05
1.00
0.95
0.90
0.85
predator CV, target 400
prey CV, target 400
predator CV, target 600
prey CV, target 600
0
10^{-2}
10^{-1}
feedback gain Kp
(d) control effort (harvest)
predator yield / tick
0.04
0.02
0.00
−0.02
−0.04
target 400
target 600
0
10^{-2}
10^{-1}
feedback gain Kp